FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 11

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Recap of previous lecture

"COMPLEX"



Topics to be covered

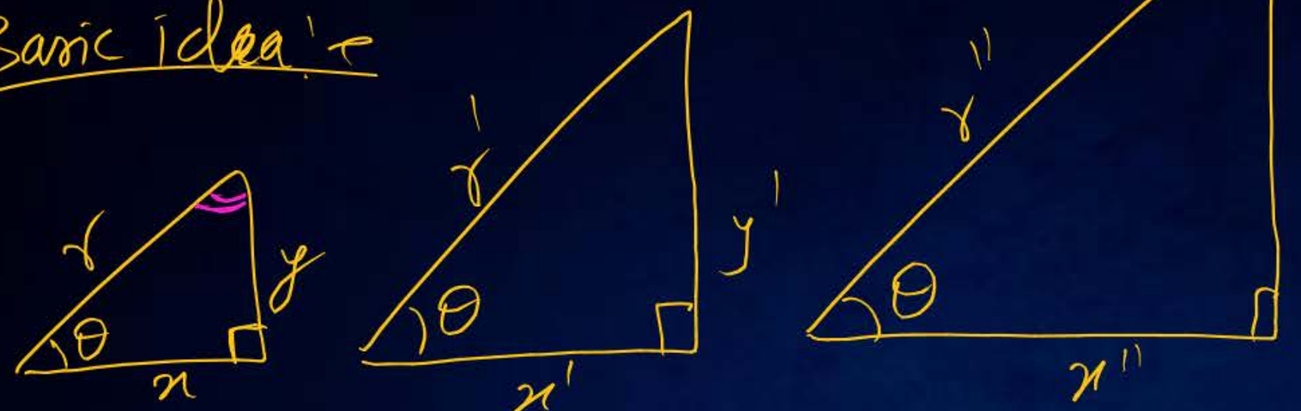
"TRIGONOMETRY"

TRI → Three Sides

GONO METRY → Measure



Basic idea :-



$\frac{y}{r} = \frac{y'}{r'} = \frac{y''}{r''} = \sin\theta$

$\frac{x}{r} = \frac{x'}{r'} = \frac{x''}{r''} = \cos\theta$

$\frac{y}{x} = \frac{y'}{x'} = \frac{y''}{x''} = \tan\theta$

x = Base

y = $\perp^{r}$ (In front of θ)

r = Hypotenuse (In front of Right Angle)



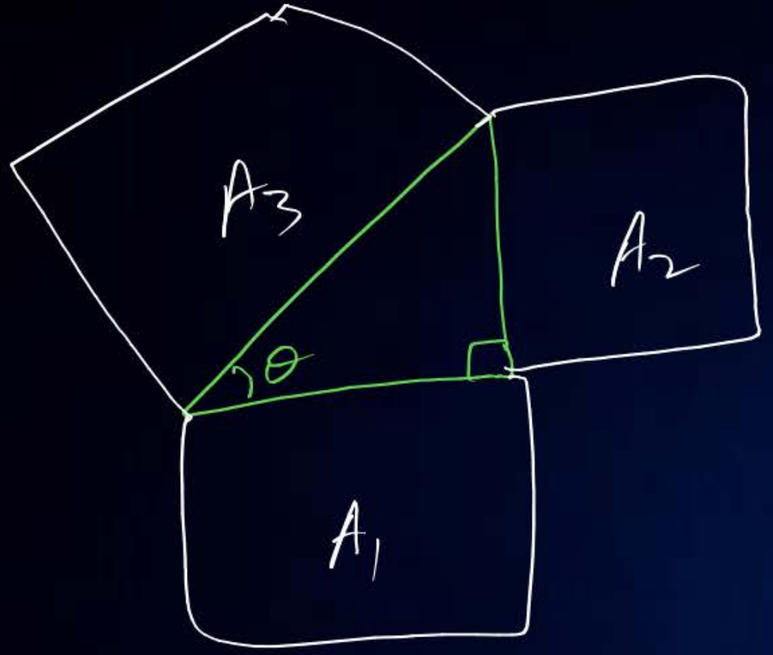
$r^2 = x^2 + y^2$

$(Hyp)^2 = (\perp^{r})^2 + (Base)^2$ → Pythagoras th.

$\text{Cosec}\theta = \frac{r}{y} = \frac{1}{\sin\theta}$

$\sec\theta = \frac{r}{x} = \frac{1}{\cos\theta}$

$\cot\theta = \frac{x}{y} = \frac{1}{\tan\theta}$



$$A_3 = A_1 + A_2$$

$$(Hyp)^2 = (Base)^2 + (\perp^r)^2$$

① $\sin\theta = \frac{\perp^r}{Hyp}$, $\cos\theta = \frac{Base}{Hyp}$, $\tan\theta = \frac{\perp^r}{Base}$

$\text{Cosec}\,\theta = \frac{Hyp}{\perp^r}$, $\sec\theta = \frac{Hyp}{Base}$, $\cot\theta = \frac{Base}{\perp^r}$

Funny: P B P / H H B

P → $\sin\theta$, B → $\cos\theta$, P → $\tan\theta$

① $\perp^r$ → Input of θ

② Hyp → ∵ ∵ Right Angle

③ $\sin\theta \neq \sin \times \theta$ while $(\sin\theta)^2 = \sin^2\theta$

④ $\sin^{-1}\theta \neq \frac{1}{\sin\theta} = \text{Cosec}\,\theta$

∵ $\sin^{-1}\theta$ & $\text{Cosec}\,\theta$ are two different functions where $\text{Cosec}\,\theta$ = Reciprocal of $\sin\theta$

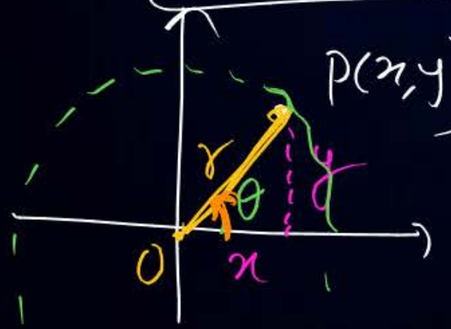
Trig Identities

(1) $\sin\theta \times \operatorname{cosec}\theta = 1$

(2) $\cos\theta \times \sec\theta = 1$

(3) $\tan\theta \times \cot\theta = 1$

(4) $\boxed{\sin^2\theta + \cos^2\theta = 1}$



$\sin\theta = \frac{y}{r}$

$\cos\theta = \frac{x}{r}$

$\tan\theta = \frac{y}{x}$

$x^2 + y^2 = r^2$

$LHS = \sin^2\theta + \cos^2\theta = \frac{y^2}{r^2} + \frac{x^2}{r^2} = \frac{x^2+y^2}{r^2} = \frac{r^2}{r^2} = 1 = RHS$

(5) $\boxed{1 + \tan^2\theta = \sec^2\theta}$ or $\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$

$LHS = 1 + \tan^2\theta = 1 + \frac{y^2}{x^2} = \frac{x^2+y^2}{x^2} = \frac{r^2}{x^2} = \left(\frac{r}{x}\right)^2 = (\sec\theta)^2 = \sec^2\theta = RHS$

(6) $\boxed{1 + \cot^2\theta = \operatorname{cosec}^2\theta}$ or $\operatorname{cosec}\theta - \cot\theta = \frac{1}{\operatorname{cosec}\theta + \cot\theta}$

$LHS = 1 + \frac{x^2}{y^2} = \frac{y^2+x^2}{y^2} = \frac{r^2}{y^2} = \left(\frac{r}{y}\right)^2 = \operatorname{cosec}^2\theta = RHS$

Here r is always +ve ie $\boxed{r > 0}$

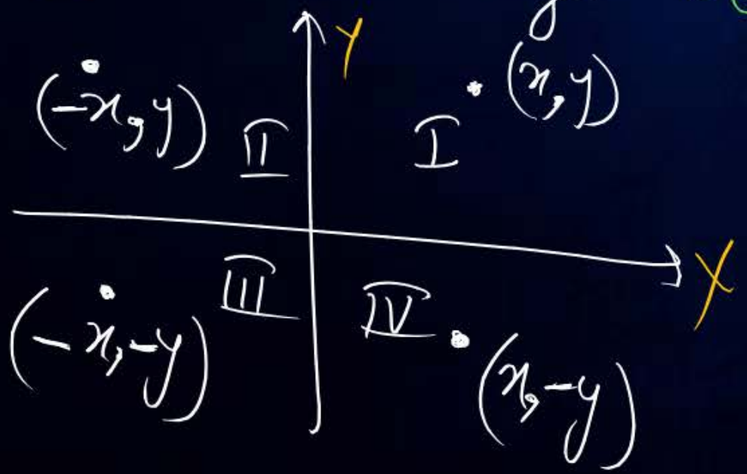
(7) $\frac{\sin\theta}{\cos\theta} = \frac{y/r}{x/r} = \frac{y}{x} = \tan\theta$

Sign of Trig Ratios →

Sine & Cosec are +ve & Rest are -ve | All +ve

Tan & Cot are +ve & Rest are -ve | Cosine & Sec are +ve & Rest are -ve

"All School To College" i.e. ASTC



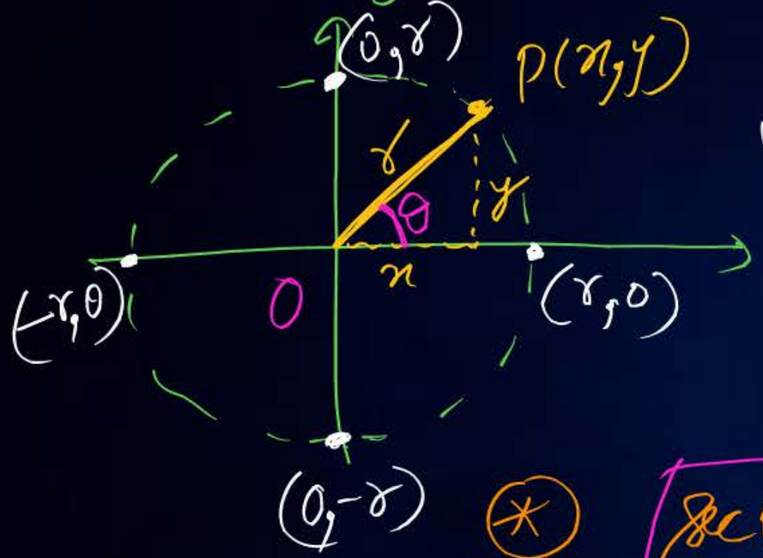
$\sin\theta = \frac{y}{r}$, $\operatorname{Cosec}\theta = \frac{r}{y}$

$\cos\theta = \frac{x}{r}$, $\sec\theta = \frac{r}{x}$

$\tan\theta = \frac{y}{x}$, $\cot\theta = \frac{x}{y}$

* Sign of $\sin\theta$ & $\operatorname{Cosec}\theta$ depends only on y ($\because r > 0$)
So these two func^n are +ve in 1st & 2nd Quadrant & are -ve in 3rd & 4th Quad
* Sign of $\cos\theta$ & $\sec\theta$ depends only on x ($\because r > 0$)
So these two func^n are +ve in 1st & 4th Quad & -ve in 2nd & 3rd Quad.
* Similarly $\tan\theta$ & $\cot\theta$ are +ve in 1st & 111rd Quad

Ranges of Trig Ratios →



$\frac{x}{r} = \cos\theta$ & $\frac{y}{r} = \sin\theta$

$\because -r \le x \le r$

$-r \le r\cos\theta \le r$

$\boxed{-1 \le \cos\theta \le 1}$

Similarly $-r \le y \le r$

$-r \le r\sin\theta \le r$

$\boxed{-1 \le \sin\theta \le 1}$

& $\boxed{\text{Cosec}\,\theta \le -1 \text{ or } \text{Cosec}\,\theta \ge 1}$

(*) $\boxed{\sec\theta \le -1 \text{ or } \sec\theta \ge 1}$

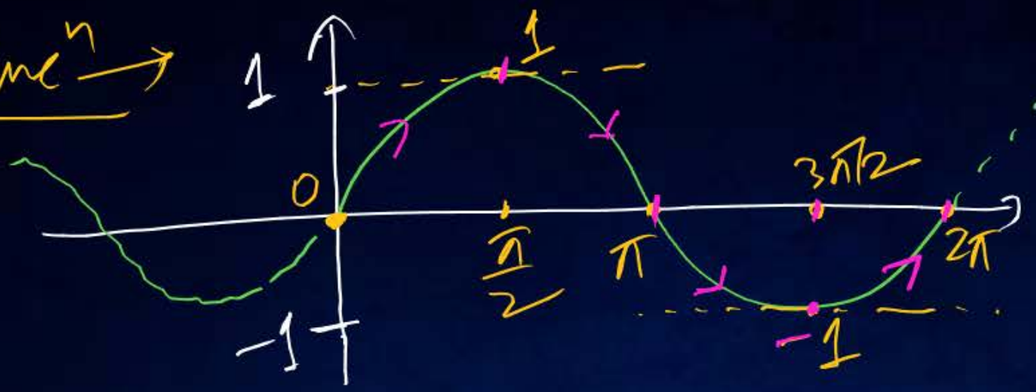
(*) $\tan\theta = \frac{y}{x}$

- At $x=0$, y = -ve then $\tan\theta = -\infty$
- At $x=0$, y = +ve then $\tan\theta = +\infty$
- At $y=0$, then $\tan\theta = 0$

$\therefore \boxed{-\infty < \tan\theta < \infty}$ & $\boxed{-\infty < \cot\theta < \infty}$

Graph of Trig fun^n →

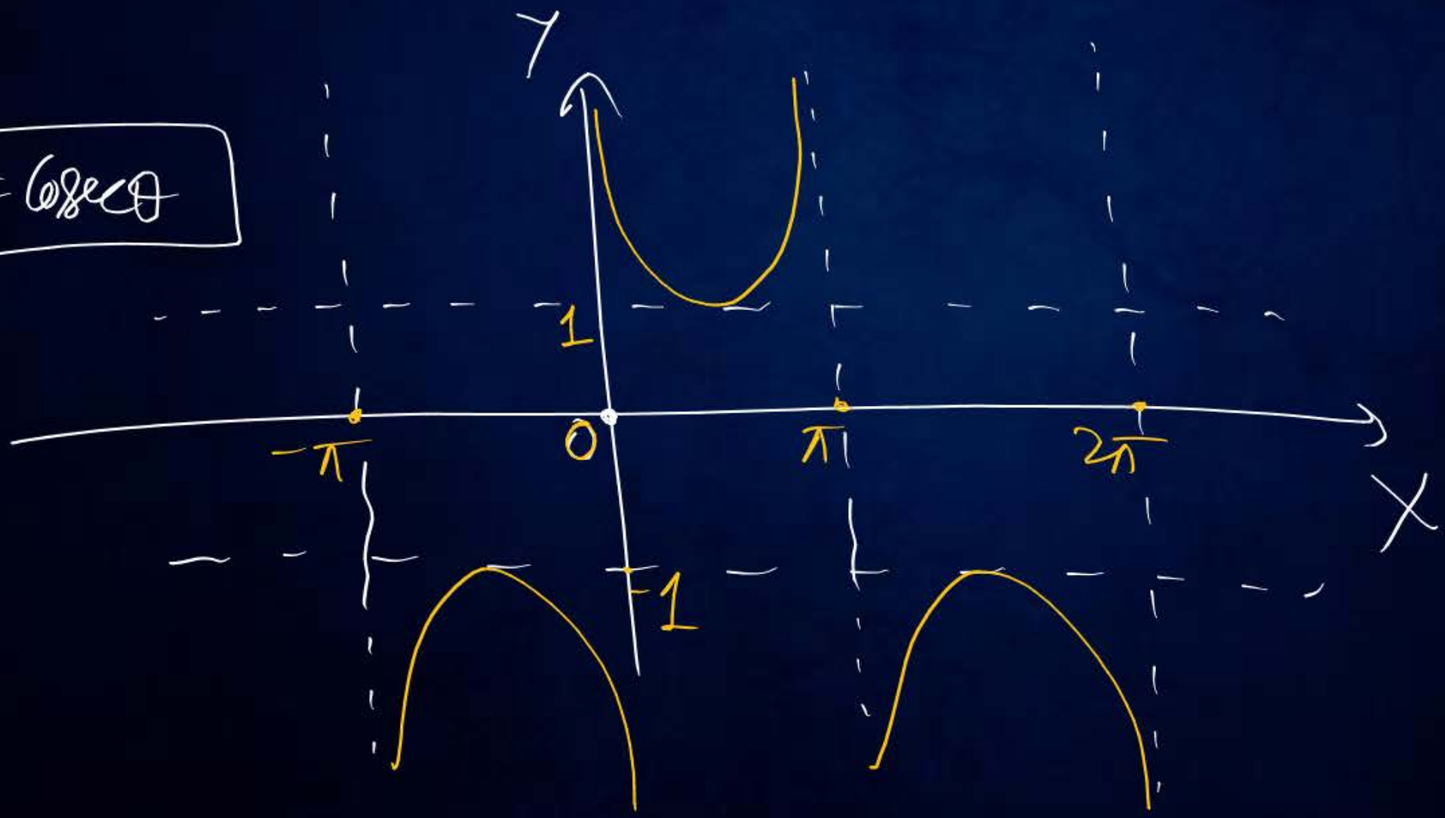
① $y = \sin x$



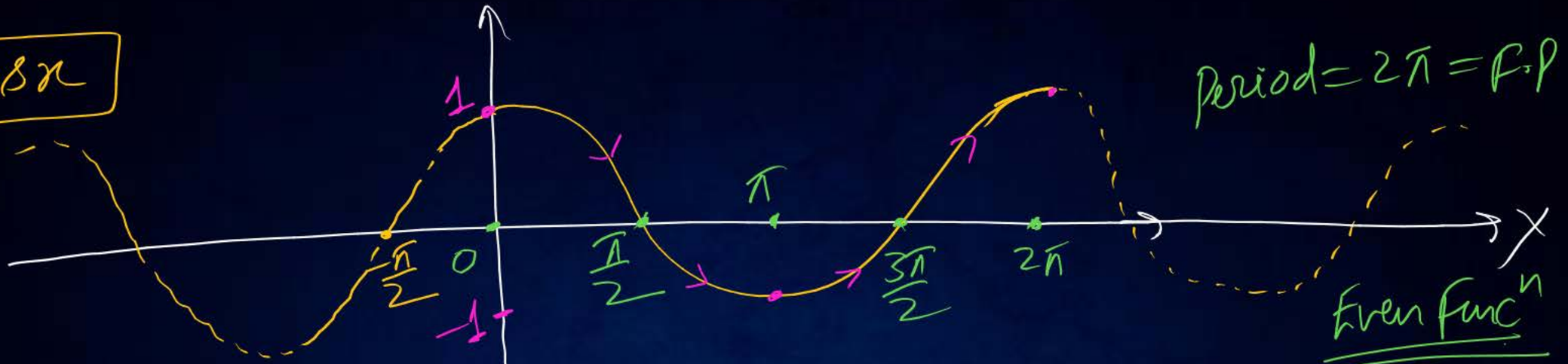
Period $= 2\pi =$ F.P.

(odd fun^n)

② $y = \csc\theta$



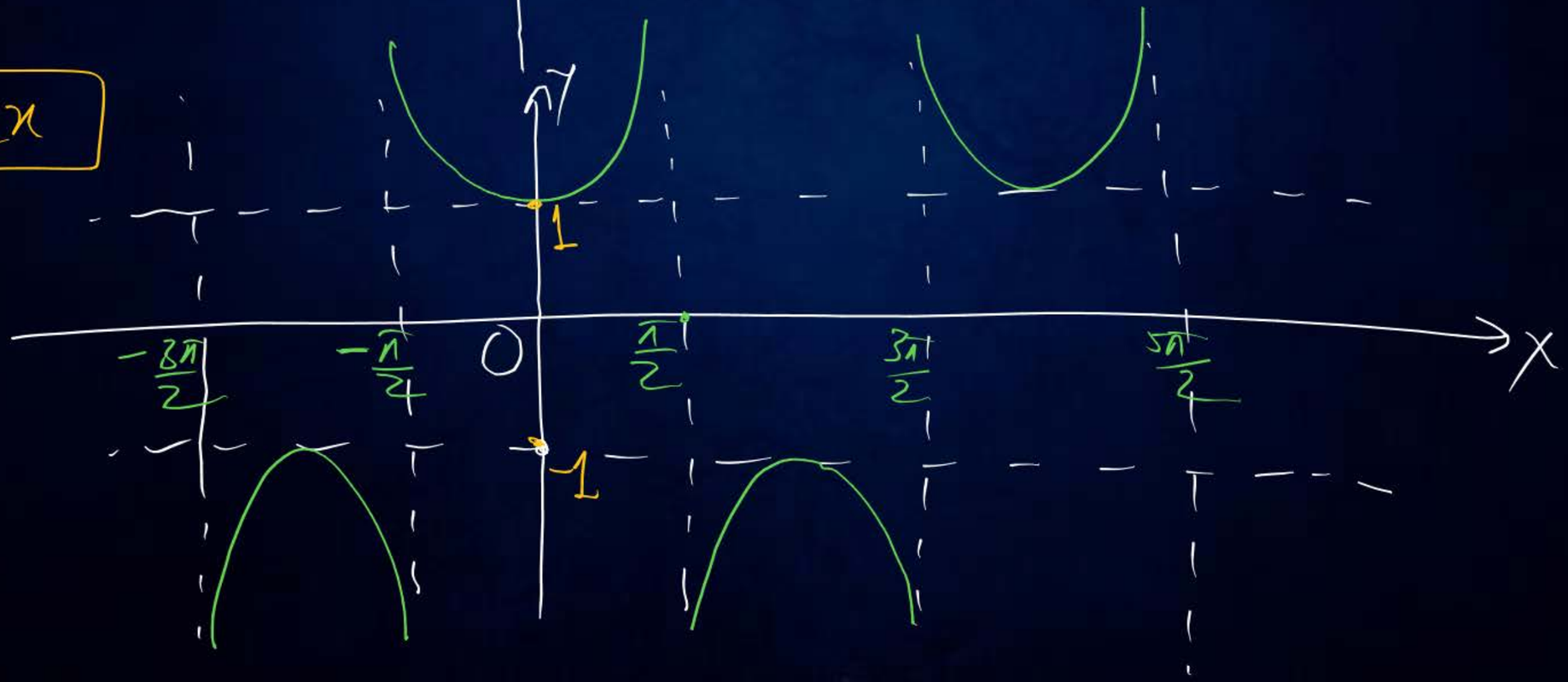
$y = \cos x$



Period $= 2\pi = F.P$

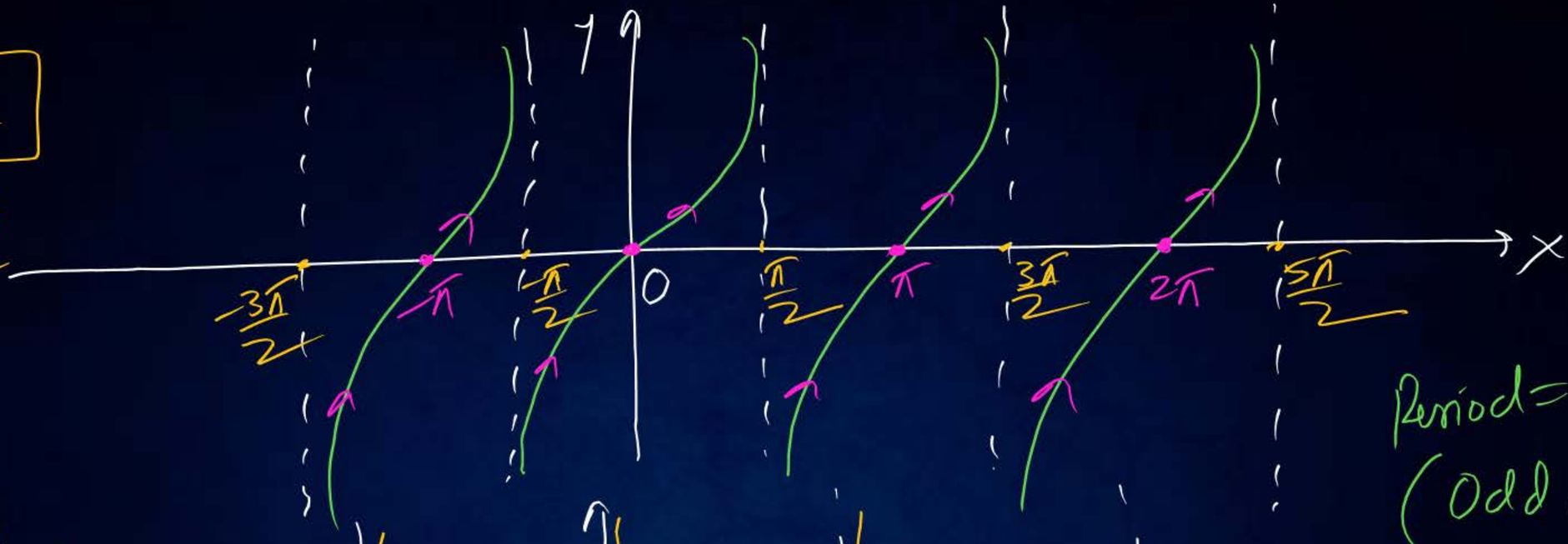
Even Func^n

$y = \sec x$



$y = \tan x$

$= \frac{\sin x}{\cos x}$



Period $= \pi =$ F.P

(Odd fun^n)

Always Increasing

$y = \cot x = \frac{\cos x}{\sin x}$

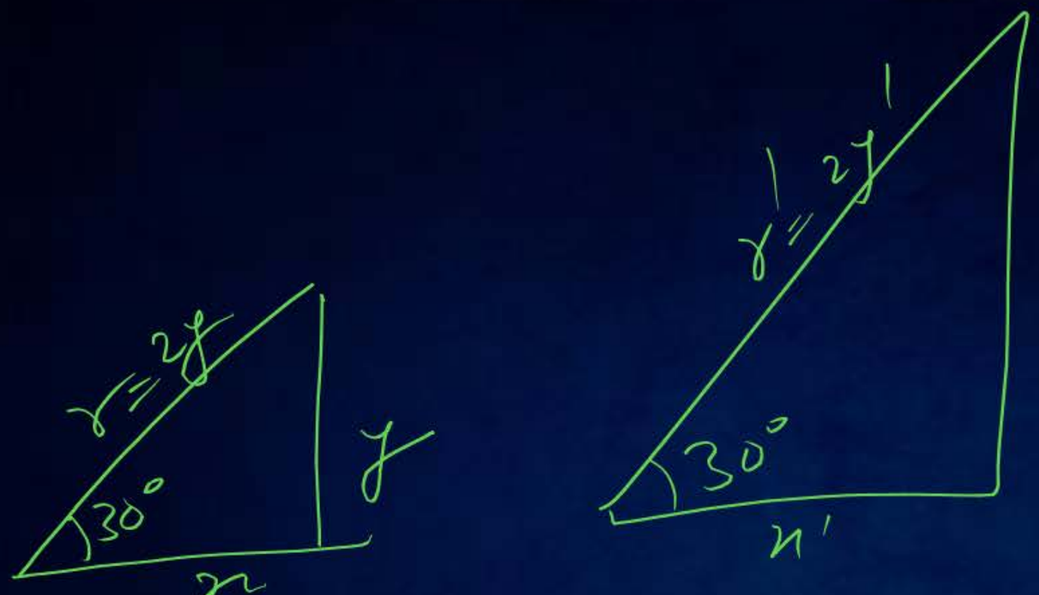


$\cot x =$ Always decreasing

Trig Ratios at Various Angles →

$\because \pi^c = 180^\circ$, $\tan\theta = \frac{\sin\theta}{\cos\theta}$

Trig Ratio \ Angle	0°	$30^\circ = \pi/6$	$45^\circ = \pi/4$	$60^\circ = \pi/3$	$90^\circ = \pi/2$	$120^\circ = 2\pi/3$	$135^\circ = 3\pi/4$	$150^\circ = 5\pi/6$	$180^\circ = \pi$
$\sin\theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\operatorname{Cosec}\theta = \frac{1}{\sin\theta}$	N.D	2	$\sqrt{2}$	$2/\sqrt{3}$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D
$\cos\theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0	$-1/2$	$-1/\sqrt{2}$	$-\sqrt{3}/2$	-1
$\sec\theta = \frac{1}{\cos\theta}$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D	-2	$-\sqrt{2}$	$-2/\sqrt{3}$	-1
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D	$-\sqrt{3}$	-1	$-1/\sqrt{3}$	0
$\cot\theta = \frac{1}{\tan\theta}$	N.D	$\sqrt{3}$	1	$1/\sqrt{3}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	N.D



$$\sin 30^\circ = \frac{y}{r} = \frac{y}{2y} = \frac{1}{2}$$

$$\sin 30^\circ = \frac{y'}{r'} = \frac{y'}{2y'} = \frac{1}{2}$$

Eg: if $\sin\theta = \frac{-12}{13}$ and θ lies in 3rd Quad. Then evaluate $\cos\theta$ & $\tan\theta$?

Sol: $\sin^2\theta + \cos^2\theta = 1$

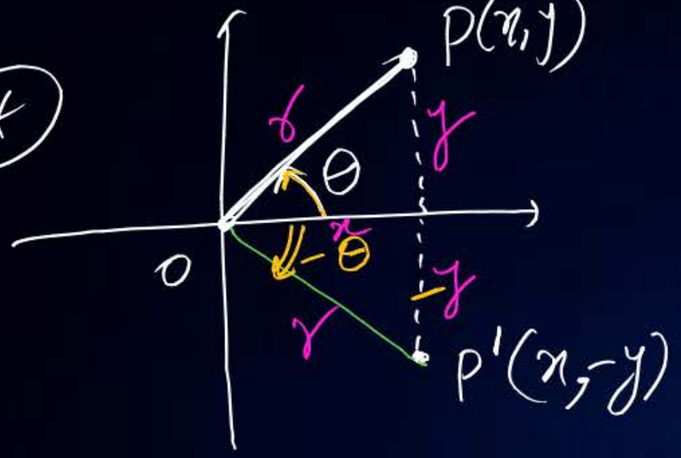
$\Rightarrow \cos^2\theta = 1 - \sin^2\theta = 1 - \left(\frac{-12}{13}\right)^2$

$\cos\theta = \pm\left(\frac{5}{13}\right) \Rightarrow \cos\theta = \frac{-5}{13}$

$\because$ in 3rd Quad, $\cos\theta$ is -ve

& $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-12/13}{-5/13} = \frac{12}{5}$

(*)



→ $\sin\theta = \frac{y}{\gamma}$ then

$\sin(-\theta) = \frac{-y}{\gamma} = -\sin\theta$

→ $\cos\theta = \frac{x}{\gamma}$ then

$\cos(-\theta) = \frac{x}{\gamma} = \cos\theta$

i.e $\sin(-\theta) = -\sin\theta$

$\operatorname{Cosec}(-\theta) = -\operatorname{Cosec}\theta$

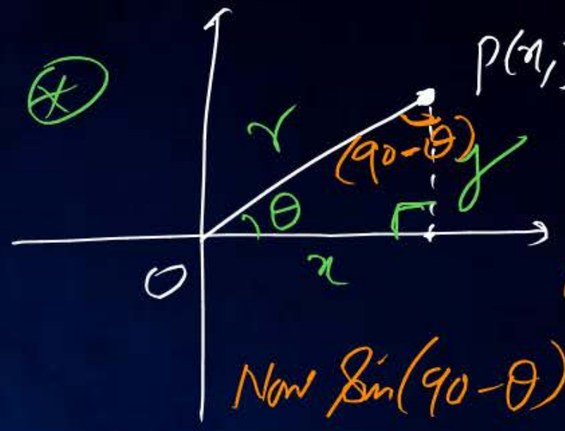
$\cos(-\theta) = \cos\theta$

$\sec(-\theta) = \sec\theta$

$\tan(-\theta) = -\tan\theta$

$\cot(-\theta) = -\cot\theta$

(*)



$\sin\theta = \frac{y}{\gamma}$, $\cos\theta = \frac{x}{\gamma}$

Now $\sin(90-\theta) = \frac{\text{Perp}}{\text{Hyp}} = \frac{x}{\gamma} = \cos\theta$

Again $\cos(90-\theta) = \frac{\text{Base}}{\text{Hyp}} = \frac{y}{\gamma} = \sin\theta$

& $\tan(90-\theta) = \frac{\text{Perp}}{\text{B}} = \frac{x}{y} = \cot\theta$

So $\cot(90-\theta) = \tan\theta$

& $\sec(90-\theta) = \operatorname{Cosec}\theta$

$\operatorname{Cosec}(90-\theta) = \sec\theta$

(8) $\sin(90+\theta) = +\cos\theta$

$\cos(90+\theta) = -\sin\theta$

$\tan(90+\theta) = -\cot\theta$

$\cot(90+\theta) = -\tan\theta$

$\sec(90+\theta) = -\operatorname{cosec}\theta$

$\operatorname{cosec}(90+\theta) = +\sec\theta$

HINT: $\sin(A+B) = \sin A\cos B + \cos A\sin B$

Put $B = \frac{\pi}{2}$, $\sin\left(\frac{\pi}{2}+A\right) = \sin A(0) + \cos A(1)$

$= \cos A$

(9) $\sin(180-\theta) = +\sin\theta$

$\operatorname{cosec}(180-\theta) = +\operatorname{cosec}\theta$

$\cos(180-\theta) = -\cos\theta$

$\sec(180-\theta) = -\sec\theta$

$\tan(180-\theta) = -\tan\theta$

$\cot(180-\theta) = -\cot\theta$

Put $A = 180$, $B = -\theta$

& then proceed

(10) $\sin(180+\theta) = -\sin\theta$

$\operatorname{cosec}(180+\theta) = -\operatorname{cosec}\theta$

$\cos(180+\theta) = -\cos\theta$

$\sec(180+\theta) = -\sec\theta$

$\tan(180+\theta) = +\tan\theta$

$\cot(180+\theta) = +\cot\theta$

Put $A = 180$, $B = \theta$

in $\sin(A+B) = ?$

(*) $\sin(360^\circ - \theta) = -\sin\theta$

$\operatorname{cosec}(360 - \theta) = -\operatorname{cosec}\theta$

$\cos(360 - \theta) = +\cos\theta$

$\sec(360 - \theta) = +\sec\theta$

$\tan(360 - \theta) = -\tan\theta$

$\cot(360 - \theta) = -\cot\theta$

(*) $\sin(360 + \theta) = +\sin\theta$

$\operatorname{cosec}(360 + \theta) = +\operatorname{cosec}\theta$

$\cos(360 + \theta) = +\cos\theta$

$\sec(360 + \theta) = +\sec\theta$

$\tan(360 + \theta) = +\tan\theta$

$\cot(360 + \theta) = +\cot\theta$

Trig. Ratios of Sum and Difference of two angles

(1) $\cos(A-B) = \cos A \cos B + \sin A \sin B$

(2) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

} Learn

(3) $\sin(A-B) = \sin A \cos B - \cos A \sin B$

(4) $\sin(A+B) = \sin A \cos B + \cos A \sin B$

(5) $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

(6) $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

(7) $\tan(A+B+C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$

Proof: (3) $\sin(A-B) = \cos(90-(A-B))$

$= \cos[(90-A)+B]$

$= \cos(90-A)\cdot\cos B - \sin(90-A)\sin B$

$= \sin A \cos B - \cos A \sin B$ Hence proved

(4) Put $B = -B$ in above Result

$\sin(A-(-B)) = \sin A \cos(-B) - \cos A \sin(-B)$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$ =

(7) $\tan(A+B+C) = \tan[(A+B)+C]$

$= \dfrac{\tan(A+B) + \tan C}{1 - \tan(A+B)\cdot\tan C}$ ..... ☺

Eg: if $\sin A = \frac{3}{5}$; $0 < A < \frac{\pi}{2}$ & $\cos B = \frac{-12}{13}$; $\pi < B < \frac{3\pi}{2}$ then evaluate $\sin(A-B)$ & $\tan(A-B) = ?$

Sol: $\cos A = \pm\sqrt{1-\sin^2 A} = \pm\sqrt{1-\frac{9}{25}} = \pm\frac{4}{5}$

$\because$ A lies in 1st Quad, so $\cos A = +\frac{4}{5}$

Now $\sin B = \pm\sqrt{1-\cos^2 B} = \pm\sqrt{1-\frac{144}{169}} = \pm\frac{5}{13}$

$\because$ B lies in 3rd Quadrant, so $\sin B = -\frac{5}{13}$

$\sin(A-B) = \sin A\cos B - \cos A \sin B = A = -\frac{16}{65}$

$\tan(A-B) = \frac{\tan A - \tan B}{1+\tan A \tan B} \Rightarrow A = \frac{16}{63}$

Eg: Evaluate $\sin\frac{7\pi}{12}\cos\frac{\pi}{4} - \cos\frac{7\pi}{12}\sin\frac{\pi}{4} = ?$

$= \sin\left(\frac{7\pi}{12} - \frac{\pi}{4}\right) = \sin\left(\frac{4\pi}{12}\right) = \sqrt{3}/2$

Q $\sin^2 6x - \sin^2 4x = ?$

Wk. that: $\sin^2 A - \sin^2 B = \sin(A+B)\cdot\sin(A-B)$

$\Rightarrow \sin^2 6x - \sin^2 4x = \sin(6x+4x)\cdot\sin(6x-4x)$

$= \sin 10x \cdot \sin 2x$

Q Evaluate $\sin 75^\circ \cos 15^\circ = ?$

$= \frac{1}{2}\left[2\sin 75^\circ \cos 15^\circ\right] = \frac{1}{2}\left[\sin(75+15) + \sin(75-15)\right]$

$= \frac{1}{2}\left[\sin 90 + \sin 60\right]$

$= \frac{1}{2}\left(1 + \frac{\sqrt{3}}{2}\right)$

(*) $\sin(A+B)\sin(A-B) = \sin^2A - \sin^2B$

& $\cos(A+B)\cos(A-B) = \cos^2A - \sin^2B$

LHS $= \sin(A+B)\cdot\sin(A-B)$

$= (\sin A\cos B + \cos A\sin B)(\sin A\cos B - \cos A\sin B)$

$= \sin^2A\cos^2B - \cos^2A\sin^2B$

$= \sin^2A(1-\sin^2B) - (1-\sin^2A)\sin^2B$

$= \sin^2A - \sin^2A\sin^2B - \sin^2B + \sin^2A\sin^2B$

$= \sin^2A - \sin^2B$

(*) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

$2\cos A\sin B = \sin(A+B) - \sin(A-B)$

$2\cos A\cos B = \cos(A+B) + \cos(A-B)$

$2\sin A\sin B = \cos(A-B) - \cos(A+B)$

Proof of 4th : $\cos(A-B) = \cos A\cos B + \sin A\sin B$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

Subtract: $\cos(A-B) - \cos(A+B) = 2\sin A\sin B$

(*) if $A+B=C$ & $A-B=D \Rightarrow A=\frac{C+D}{2}$ & $B=\frac{C-D}{2}$

(1) $\because \sin(A+B)+\sin(A-B)=2\sin A\cos B$

$\sin C+\sin D=2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$

(2) $\because \sin(A+B)-\sin(A-B)=2\cos A\sin B$

$\sin C-\sin D=2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$

(3) $\because \cos(A+B)+\cos(A-B)=2\cos A\cos B$

ie $\cos C+\cos D=2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$

(4) $\cos(A-B)-\cos(A+B)=2\sin A\sin B$

ie $\cos D-\cos C=2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

or $\cos C-\cos D=2\sin\frac{C+D}{2}\sin\left(\frac{D-C}{2}\right)$

(*) $\sin 2A = 2\sin A\cos A = \dfrac{2\tan A}{1+\tan^2 A}$, $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \frac{1-\tan^2 A}{1+\tan^2 A}$$

$\Rightarrow \dfrac{1+\cos 2A}{2} = \cos^2 A$

$\Rightarrow \dfrac{1-\cos 2A}{2} = \sin^2 A$

$\rightarrow \sin(A+B) = \sin A\cos B + \cos A\sin B$

Put $B = A$, $\sin(A+A) = \sin A\cos A + \cos A\sin A$

$\sin 2A = 2\sin A\cos A$

$\rightarrow \cos(A+B) = \cos A\cos B - \sin A\sin B$

Put $B = A$, $\cos(A+A) = \cos A\cos A - \sin A\sin A$

$\cos 2A = \cos^2 A - \sin^2 A$

$\rightarrow \tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

Put $B = A$

(*) $\sin 3A = 3\sin A - 4\sin^3 A$

$\cos 3A = 4\cos^3 A - 3\cos A$

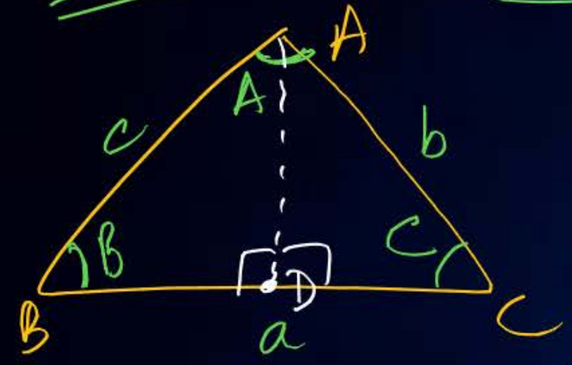
$\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Put $B = 2A$ in above Results

eg Evaluate $8\cos^3\left(\frac{\pi}{9}\right) - 6\cos\left(\frac{\pi}{9}\right) = ?$

$= 2\left[4\cos^3\left(\frac{\pi}{9}\right) - 3\cos\left(\frac{\pi}{9}\right)\right] = 2\left[\cos\left(3\times\frac{\pi}{9}\right)\right] = 2\left(\frac{1}{2}\right) = 1$

Note: $\angle A+\angle B+\angle C=180^\circ$, $a+b>c$, $b+c>a$, $c+a>b$



SINE Rule: $\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$

$\sin B=\frac{AD}{AB}\Rightarrow AD=c\sin B$

again $\sin C=\frac{AD}{AC}\Rightarrow AD=b\sin C$

$\therefore c\sin B=b\sin C\Rightarrow \frac{c}{\sin C}=\frac{b}{\sin B}$

Projection formulae:

$a=b\cos C+c\cos B$

$b=a\cos C+c\cos A$

$c=a\cos B+b\cos A$

"Any side of a $\triangle$ is equal to sum of projections of other two sides on it"

Proof: BD = Projection of line AB on line BC

$\cos B=\frac{BD}{AB}\Rightarrow BD=c\cos B$

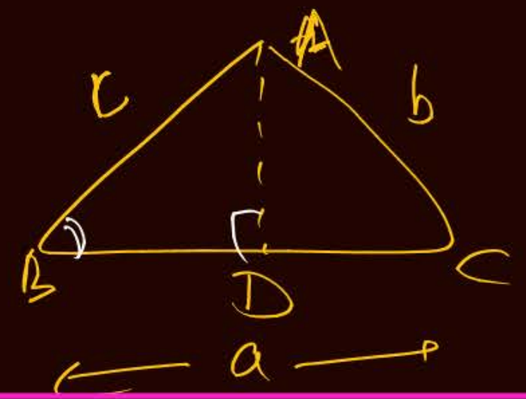
DC = Projection of line AC on line BC

$\cos C=\frac{DC}{AC}\Rightarrow DC=b\cos C$

Hence $BC=BD+DC$

$a=c\cos B+b\cos C$

Area of $\triangle ABC$ –



$$\text{Area} = \frac{1}{2}bc\sin A = \frac{1}{2}ab\sin C = \frac{1}{2}ca\sin B$$

Proof: $\frac{AD}{AB} = \sin B \Rightarrow AD = c\sin B$

New Area of $\triangle ABC = \frac{1}{2} \times \text{Base} \times \text{Height}$

$$= \frac{1}{2}(a) \times (c\sin B)$$

$$= \frac{1}{2}ac\sin B$$

COSINE Rule –

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$\cos B = \frac{a^2+c^2-b^2}{2ac}$$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

eg In a $\triangle ABC$, $a=3, b=5, c=7$ then Find $\cos A, \cos B, \cos C$

$\frac{13}{14}$ $\quad \frac{11}{14}$ / $\frac{-1}{2}$

Q: In a $\triangle ABC$ if $\angle A = 60^\circ$ and $b = 4$, $c = \sqrt{3}$ then find area

Sol: $\text{Area} = \frac{1}{2} bc \sin A = \frac{1}{2} \times 4\sqrt{3} \times \sin 60^\circ$

$= \frac{1}{2} (4\sqrt{3}) \left(\frac{\sqrt{3}}{2}\right) = 3$

Q: In a $\triangle ABC$, $a = 2$, $b = 3$, $\sin A = \frac{2}{3}$ then $\angle B = ?$

Sol: $\frac{a}{\sin A} = \frac{b}{\sin B}$

$\frac{2}{2/3} = \frac{3}{\sin B} \Rightarrow \sin B = 1 \Rightarrow \angle B = 90^\circ$

Trig Eqn^n

(1) $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$

(2) $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$

(3) $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$

$\Big\}\ n \in I$

Proof (3) $\rightarrow \frac{\sin\theta}{\cos\theta} = \frac{\sin\alpha}{\cos\alpha} \Rightarrow \sin\theta\cos\alpha = \cos\theta\sin\alpha$

$\sin\theta\cos\alpha - \cos\theta\sin\alpha = 0$

$\sin(\theta - \alpha) = 0$

$\theta - \alpha = n\pi$

$\boxed{\theta = n\pi + \alpha}$

Note ①

$\sin\pi = 0$

$\sin 2\pi = 0$

$\sin 3\pi = 0$

$\vdots$

$\sin(-\pi) = 0$

$\sin(-2\pi) = 0$

$\Downarrow$

$\boxed{\sin n\pi = 0}$

② $\cos 0 = 1$

$\cos\pi = -1$

$\cos 2\pi = 1$

$\cos 3\pi = -1$

$\cos 4\pi = 1$

$\cos 5\pi = -1$

$\vdots$

$\Downarrow$

$\boxed{\cos n\pi = (-1)^n}$

③ $\sin\theta = 0 \Rightarrow \theta = n\pi$

$\cos\theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$

$\tan\theta = 0 \Rightarrow \theta = n\pi$

, $n \in I$

④ $\sec\theta = 0 \Rightarrow \theta =$ No Sol.

& $\operatorname{cosec}\theta = 0 \Rightarrow \theta =$ No Sol.

$\therefore$ $\sec\theta \le -1$ or $\sec\theta \ge 1$

& $\operatorname{cosec}\theta \le -1$ or $\operatorname{cosec}\theta \ge 1$

(*) $\left.\begin{array}{l} \sin^2\theta = \sin^2\alpha \\ \cos^2\theta = \cos^2\alpha \\ \tan^2\theta = \tan^2\alpha \end{array}\right\} \Rightarrow \boxed{\theta = n\pi \pm \alpha}$

Proof: $\sin^2\theta = \sin^2\alpha$

$2\sin^2\theta = 2\sin^2\alpha$

$1 - \cos 2\theta = 1 - \cos 2\alpha$

$\cos 2\theta = \cos 2\alpha$

$2\theta = 2n\pi \pm 2\alpha$

$\boxed{\theta = n\pi \pm \alpha}$

eg: solve $2\sin^2 x - 1 = 0$

$\sin^2 x = \frac{1}{2} = \left(\frac{1}{\sqrt{2}}\right)^2$

$\sin^2 x = \sin^2 \frac{\pi}{4}$

$x = n\pi \pm \frac{\pi}{4}$

eg: $\cos\theta = \frac{1}{2}$ then $\theta = ?$

$\cos\theta = \cos\frac{\pi}{3}$

$\theta = 2n\pi \pm \left(\frac{\pi}{3}\right)$

Q: solve:

$\tan\theta + \tan 2\theta + \tan 3\theta = \tan\theta \tan 2\theta \tan 3\theta$

$\tan\theta + \tan 2\theta = (\tan\theta \tan 2\theta - 1)\tan 3\theta$

$\frac{\tan\theta + \tan 2\theta}{1 - \tan\theta \tan 2\theta} = -\tan 3\theta$

$\tan(\theta + 2\theta) = -\tan 3\theta$

$2\tan 3\theta = 0 \Rightarrow \tan 3\theta = 0$

$3\theta = n\pi \Rightarrow \theta = \frac{n\pi}{3}$


Thank
YOU